Choosing a Spectral Prediction Target for EEG: An Idealized Analysis of Five Classical Frequency Bands

Abstract

This note develops a theoretical rationale for masked prediction of EEG band power rather than sample-wise waveform reconstruction. The target is defined using five *ideal, non-overlapping classical frequency bands*: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and γ (30–45 Hz). In an idealized Gaussian spectral model, Fourier phases are ancillary for a state that changes only the power spectrum. If the spectrum is constant within each band, the five band-power statistics are sufficient for the five spectral levels; averaging within a band reduces periodogram noise, and the logarithm makes the target's variance independent of those levels. A conditional-risk decomposition then identifies the state-dependent and irreducible components of masked prediction. These are statements about an explicitly specified model, not a proof that a particular EEG encoder or training objective must outperform waveform reconstruction in practice.

1 Question and target

A masked EEG encoder receives a context C containing visible channel–time patches and predicts a property of a masked patch x_M. Which property should it predict? For a downstream label that depends primarily on sustained rhythmic activity, a natural candidate is the patch's spectral energy in a small number of physiologically meaningful bands. This choice removes the requirement to reproduce the precise sample-by-sample realization of the masked waveform. It does *not* assert that all EEG labels are determined by band power: transient morphology, spatial relationships, and phase coupling can also be essential.

1.1 Why power? Rhythmic strength without exact phase

Consider first a single oscillation, $x(t) = A\cos(2\pi ft + \phi)$. Its mean square over a window containing an integer number $q \geq 1$ of periods, $T = q/f$, is

$$\begin{aligned}\frac{1}{T}\int_0^T x(t)^2\,dt &= \frac{A^2}{T}\int_0^T \cos^2(2\pi ft+\phi)\,dt \\ &= \frac{A^2}{2} + \frac{A^2}{2T}\int_0^T \cos(4\pi ft + 2\phi)\,dt = \frac{A^2}{2}. \end{aligned} \tag{1}$$

Thus mean-square power measures the oscillation's squared amplitude but does not depend on where its peak happens to fall within a complete-cycle window. Squaring alone does *not* remove phase: $x(t)^2$ still oscillates and still depends on ϕ. Phase invariance comes from averaging over complete periods, or, for an ideal frequency-domain target, from taking squared Fourier magnitudes. For a finite window containing a non-integer number of periods, the boundary term need not vanish. This matters especially for slow rhythms in short patches.

For a multi-frequency signal, ideal band-pass filtering isolates a classical rhythm range. Parseval's identity equates the filtered signal's mean square with the spectral energy in that range; the precise finite-DFT normalization appears in Eq. (9). This is why the power in, for example, the α or δ band is a natural measure of that rhythm's strength. Power does not

preserve the phase relations among frequency components or channels, which may themselves carry information for other tasks.

The quantity expected to persist across neighbouring patches is the *state-dependent spectral level*, not each patch's exact measured power. If a state lasts longer than one patch, visible patches can provide information about the expected energy of a masked patch. Its realized power still fluctuates, and a stable oscillator may have partly predictable phase across adjacent patches. The formal shared-state and conditional-independence conditions needed for the masked-prediction argument are stated separately in Assumption 2; they are not consequences of Eq. (1).

1.2 Why take the logarithm?

Power is nonnegative and often spans a wide range. For positive values, a logarithm compresses this range and makes a difference in log power approximately a *relative* difference in power:

$$\log(P + \Delta P) - \log P \approx \frac{\Delta P}{P} \qquad (|\Delta P| \ll P). \tag{2}$$

If amplitude is multiplied by $g > 0$, power is multiplied by g^2, and log power is shifted by $2 \log g$. This turns a multiplicative factor into an additive one; it does not by itself remove the factor. A fixed reference power or normalization makes the argument of the logarithm dimensionless.

There is also a statistical reason. Individual periodogram powers fluctuate roughly in proportion to their spectral level. Under the ideal model developed below, band-averaged power has conditional variance proportional to the *square* of that level, whereas log band power has conditional variance that depends only on the number of frequency bins (Proposition 3). Thus logarithmic power is a natural target when squared error should not be dominated solely by the bands or states with the largest absolute power. This argument applies exactly to $\log P$ in that model. An offset transform such as $\log(a + P)$ behaves differently at low power, as Eq. (13) will show.

We adopt the following conventional, half-open positive-frequency bands throughout:

$$\begin{aligned} &B_\delta = [0.5, 4), \quad B_\theta = [4, 8), \qquad B_\alpha = [8, 13), \\ &B_\beta = [13, 30), \quad B_\gamma = [30, 45] \quad \text{Hz}. \end{aligned} \tag{3}$$

The exact boundary convention is stipulated for the analysis; physiological rhythms do not have sharp boundaries. In particular, the word "classical" refers to these five *ideal rectangular bands*, not to five generic or overlapping spectral filters.

For one real-valued, mean-zero patch $x = (x_0, \ldots, x_{N-1})$ sampled at f_s, let

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N}, \qquad I_k = \frac{|X_k|^2}{N}, \qquad f_k = \frac{k f_s}{N}. \tag{4}$$

We use positive-frequency bins $1 \le k \le \lfloor (N-1)/2 \rfloor$, excluding DC and, when present, Nyquist. For $\mathcal{K}_b = \{k : f_k \in B_b\}$ and $m_b = |\mathcal{K}_b| > 0$, define the *mean band periodogram*

$$P_b = \frac{1}{m_b} \sum_{k \in \mathcal{K}_b} I_k, \qquad Y_b = \log P_b. \tag{5}$$

The vector $(Y_\delta, Y_\theta, Y_\alpha, Y_\beta, Y_\gamma)$ is the idealized five-dimensional prediction target. Since the Gaussian model below has $P_b > 0$ almost surely, its logarithm is well-defined once power is expressed relative to a fixed reference unit. Such a reference adds only a fixed offset to Y_b and does not alter the results below.

For illustration, at $N = 200$ and $f_s = 256$ Hz, the spacing is $\Delta f = 1.28$ Hz. The number of represented positive-frequency bins is small in the slow bands: There is no bin between 0.5 and 1.28 Hz in this example. Calling the first statistic "δ power" therefore does not imply that a short patch resolves the entire 0.5–4 Hz band. Longer windows or a different estimator are required for finer low-frequency resolution.

Table 1: Classical bands and represented DFT bins for a 200-sample patch at 256 Hz. The last column is the theoretical standard deviation of $\log P_b$ under Assumption 1; it is not an empirical EEG measurement.

Band	Range (Hz)	m_b	$\sqrt{\psi_1(m_b)}$
δ	$[0.5, 4)$	3	0.63
θ	$[4, 8)$	3	0.63
α	$[8, 13)$	4	0.53
β	$[13, 30)$	13	0.28
γ	$[30, 45]$	12	0.29

2 An idealized spectral model

The following finite-dimensional model is chosen so that its conclusions are exact within the stated observation space. For a weakly stationary Gaussian process on a long window, a Fourier-diagonal likelihood is instead an approximation, commonly associated with the Whittle likelihood. Nothing below assumes that real EEG is exactly stationary or Gaussian.

Assumption 1 (Five-band Gaussian spectral state)**.** *Let Θ denote a latent state. Conditional on $\Theta = \theta$, the positive-frequency Fourier coefficients of a real, mean-zero patch in the five bands are independent and*

$$X_k \mid \theta \sim \mathcal{CN}(0, N s_b(\theta)) \quad \text{for } k \in \mathcal{K}_b, \qquad s_b(\theta) > 0. \tag{6}$$

The negative-frequency coefficients are their complex conjugates. The observation for the sufficiency statement consists only of coefficients in $\bigcup_b \mathcal{K}_b$; any omitted frequencies are either unobserved or have a distribution independent of θ. The state affects this observation only through $\boldsymbol{s}(\theta) = (s_\delta, s_\theta, s_\alpha, s_\beta, s_\gamma)$.

This is a *piecewise-flat, ideal rectangular-band* model. In a continuous-time interpretation, s_b describes the spectral level within band B_b. An actual spectrum may vary within a band and may change smoothly at its boundaries. In that case, the exact sufficiency result below no longer follows.

2.1 Phase carries no state information in the model

Write $X_k = R_k e^{i\Phi_k}$. Circular symmetry of the complex normal in Eq. (6) gives

$$I_k \mid \theta \sim s_b(\theta)\,\mathrm{Exp}(1), \qquad \Phi_k \mid \theta \sim \mathrm{Unif}[0, 2\pi), \tag{7}$$

with independent magnitudes and phases. The phase distribution does not depend on θ.

Proposition 1 (Ancillarity of phase)**.** *Under Assumption 1, the phase vector of the observed positive-frequency coefficients is ancillary for $\boldsymbol{s}(\theta)$. The likelihood for θ depends on the observation only through the periodogram values $\{I_k\}$.*

Proof. Each conditional complex-normal density is proportional to $s_b(\theta)^{-1} \exp[-I_k/s_b(\theta)]$. In polar coordinates, its angular factor is constant. Multiplication across the observed bins leaves no phase term that depends on θ. □

This statement does *not* say that Fourier phase is always useless in EEG. It says that phase is uninformative about a state *defined to act only through the five spectral levels* in this model. Nor does it say that every aspect of amplitude is predictable: each I_k retains an exponential fluctuation even if θ is known.

2.2 Why band averaging is sufficient under piecewise flatness

Proposition 2 (Band-power sufficiency and estimation)**.** *Under Assumption 1,* $(P_b)_{b\in\{\delta,\theta,\alpha,\beta,\gamma\}}$ *is sufficient for* $\boldsymbol{s}$ *in the observed, band-limited Fourier data. For each band, the maximum-likelihood estimate is* $\widehat{s}_b = P_b$*; by invariance of maximum likelihood,* $\log P_b$ *estimates* $\log s_b$*.*

Proof. Ignoring terms independent of $\boldsymbol{s}$, the negative log-likelihood is

$$-\log p(\{X_k\} \mid \boldsymbol{s}) = \sum_b \left(m_b \log s_b + \frac{m_b P_b}{s_b} \right) + \text{const.} \tag{8}$$

The factorization criterion gives sufficiency. Differentiating the bth term with respect to s_b yields $m_b/s_b - m_b P_b/s_b^2 = 0$, so its positive minimizer is $s_b = P_b$. □

The frequency-domain mean P_b is proportional to the time-domain mean square of an *ideal circular band-pass* signal. With the DFT normalization in Eq. (4), Parseval's identity gives

$$\frac{1}{N}\sum_{n=0}^{N-1} |x_n^{(b)}|^2 = \frac{2}{N^2}\sum_{k\in\mathcal{K}_b} |X_k|^2 = \frac{2m_b}{N} P_b. \tag{9}$$

The two band-power conventions differ only by the known factor $2m_b/N$. Their logarithms differ by a band-dependent constant. This equivalence presumes the stated ideal frequency mask and excludes DC and Nyquist.

Remark 1. *Sufficiency holds for the five parameters* $\boldsymbol{s}$ within the model and the specified observation space*. If the spectral shape inside a band, the power outside the five bands, cross-channel relationships, or transient morphology also varies with the state, five band averages need not retain all state information.*

2.3 Log power and estimator noise

Proposition 3 (Distribution and variance stabilization)**.** *Under Assumption 1,*

$$P_b = s_b G_b, \qquad G_b \sim \text{Gamma}(m_b, \textit{scale} = 1/m_b). \tag{10}$$

Writing ψ *and* ψ_1 *for the digamma and trigamma functions, respectively,*

$$\mathbb{E}[\log P_b \mid \theta] = \log s_b(\theta) + \psi(m_b) - \log m_b, \tag{11}$$

$$\text{Var}(\log P_b \mid \theta) = \psi_1(m_b). \tag{12}$$

Thus the conditional noise variance on the log scale depends on the number of frequency bins, but not on the state's spectral level.

Proof. The sum of m_b independent unit-mean exponentials is Gamma with shape m_b and unit scale. Dividing by m_b gives Eq. (10). The standard log-Gamma moments are $\mathbb{E}\log G_b = \psi(m_b) - \log m_b$ and $\text{Var}(\log G_b) = \psi_1(m_b)$. □

For comparison, a single periodogram ordinate has $\text{sd}(\log I_k \mid \theta) = \pi/\sqrt{6} \approx 1.28$. Band averaging lowers this fluctuation as m_b grows, with $\psi_1(m_b) \approx 1/m_b$ for large m_b. The benefit is limited in the slow bands of short patches, where m_b is only three in Table 1. The transformed target is not noise-free: G_b is a genuine stochastic realization that remains after conditioning on the state.

An additive offset inside the logarithm changes the statement. For the mathematical family $Y_b^{(a)} = \log(a + P_b)$ with $a > 0$, a first-order delta-method approximation around $G_b = 1$ gives

$$\text{Var}(Y_b^{(a)} \mid \theta) \approx \frac{1}{m_b}\left(\frac{s_b(\theta)}{a + s_b(\theta)}\right)^2. \tag{13}$$

Unlike Eq. (12), this variance depends on the state unless the spectral level is large relative to a. Equation (13) is an approximation about a mathematical alternative, not part of the exact sufficiency argument.

3 Masked prediction and irreducible uncertainty

The sufficient-statistic result concerns inference from an *observed* patch. Masked prediction asks a different question: what can a model infer about a *missing* patch from its context? A separate temporal assumption is required.

Assumption 2 (Shared state and conditional independence)**.** *The visible context C and the masked target Y are associated with a latent state Θ. Conditional on Θ, Y is independent of C. The target has a finite second moment.*

This is an idealization of state persistence across patches. It is not implied by Assumption 1; adjacent samples of a continuous process may remain correlated even after its broad state is fixed.

Proposition 4 (Bayes risk decomposition)**.** *Let $\boldsymbol{Y}$ be any vector target satisfying Assumption 2, and put $\boldsymbol{\mu}(\theta) = \mathbb{E}[\boldsymbol{Y} \mid \Theta = \theta]$ and $\Sigma_Y(\theta) = \mathrm{Cov}(\boldsymbol{Y} \mid \Theta = \theta)$. Under squared-error loss, the Bayes predictor is $\widehat{\boldsymbol{Y}}(C) = \mathbb{E}[\boldsymbol{Y} \mid C]$ and its risk is*

$$\mathbb{E}\|\boldsymbol{Y} - \widehat{\boldsymbol{Y}}(C)\|_2^2 = \mathbb{E}\ \mathrm{tr}[\mathrm{Cov}(\boldsymbol{\mu}(\Theta) \mid C)] + \mathbb{E}\ \mathrm{tr}[\Sigma_Y(\Theta)]\,. \tag{14}$$

Proof. Conditional independence gives $\mathbb{E}[\boldsymbol{Y} \mid C] = \mathbb{E}[\boldsymbol{\mu}(\Theta) \mid C]$. The law of total covariance, conditional on C, yields $\mathrm{Cov}(\boldsymbol{Y} \mid C) = \mathrm{Cov}(\boldsymbol{\mu}(\Theta) \mid C) + \mathbb{E}[\Sigma_Y(\Theta) \mid C]$. Take traces and expectations. □

For the five-band log-power target in Eq. (5), if the conditional distribution in Assumption 1 applies to the masked patch, Eq. (11) gives

$$\mu_b(\theta) = \log s_b(\theta) + \kappa_b, \qquad \kappa_b = \psi(m_b) - \log m_b, \tag{15}$$

and Eq. (12) gives conditional variance $\psi_1(m_b)$. Consequently, the state-independent part of the five-dimensional squared-error risk is $\sum_b \psi_1(m_b)$ before any optional averaging over bands. The other term is posterior uncertainty about the five log spectral levels that remains even for the Bayes predictor with this fixed context. It is not reducible merely by improving a predictor while keeping the same context. For any predictor f, the actual excess risk is

$$R(f) - R^* = \mathbb{E}\|f(C) - \mathbb{E}[\boldsymbol{Y} \mid C]\|_2^2, \tag{16}$$

where R^* is the full Bayes risk in Eq. (14). If one instead uses $\log(a + P_b)$, the conditional sampling term generally depends on the state, as Eq. (13) illustrates.

For waveform reconstruction, suppose that, conditional on a known θ, the masked sample vector $\boldsymbol{x}_M$ and an observed waveform context $\boldsymbol{x}_C$ are jointly zero-mean Gaussian with covariance blocks $\Sigma_{MM}, \Sigma_{MC}, \Sigma_{CM}, \Sigma_{CC}$, with Σ_{CC} invertible. Then

$$\mathbb{E}[\boldsymbol{x}_M \mid \boldsymbol{x}_C, \theta] = \Sigma_{MC}\Sigma_{CC}^{-1}\boldsymbol{x}_C, \tag{17}$$

$$\mathrm{Cov}(\boldsymbol{x}_M \mid \boldsymbol{x}_C, \theta) = \Sigma_{MM} - \Sigma_{MC}\Sigma_{CC}^{-1}\Sigma_{CM}. \tag{18}$$

The trace of Eq. (18), divided by the number of masked samples, is the conditional mean-squared *waveform innovation* even when the state is known. It includes unpredictable amplitude and phase, not phase alone. Uncertainty about Θ can add further error to prediction from C.

Equations (14)–(18) motivate the hypothesis that a band-power target can direct learning toward persistent spectral-state information rather than the full waveform realization. They do *not* establish a universal ordering between the two objectives: their target dimensions, units, loss normalizations, and Bayes-risk floors differ. Nor do they prove better downstream classification, faster optimization, or greater data efficiency. Such claims require controlled empirical comparisons.

4 Where the model ceases to apply

Several EEG properties lie outside this five-band account. Sleep spindles, K-complexes, epileptiform events, and other transients may be distinguished by shape and timing rather than average energy. Cross-channel coherence, interregional phase relationships, phase–amplitude coupling, and fine spectral structure are absent from five single-channel band powers. Eye movement, muscle activity, and recording artifacts can contribute energy inside the same bands as cortical rhythms. A state transition inside a patch violates the locally persistent-state picture. Finally, finite windows limit frequency resolution, especially for the lowest frequencies, as Table 1 shows.

These are not minor technical exceptions. They delimit when spectral energy is a plausible target for learning task-relevant structure and when it may discard information that the downstream label actually needs.

5 Conclusion

Under an explicitly defined five-band Gaussian spectral model, the phases of the observed Fourier coefficients are ancillary, the five classical band powers are sufficient for the five spectral levels, and the log-power target has state-independent conditional variance. A separate shared-state assumption connects these properties to masked prediction through an exact Bayes-risk decomposition. The analysis provides a principled reason to test band-power prediction for labels tied to sustained EEG rhythms, while leaving its comparative representation quality to experiment.